Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/03/2005
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Project Title:	Checkpoint Reduced Cell Sensitivity to High Energy Particles-Induced Killing		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
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Contact Monitor:		**Contact Phone:**	
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This application is in response to the NASA Program NRA 02-OBPR-02 and addresses research related to both Bioastronautics and Fundamental Biology. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short term effect) and carcinogenesis (long term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes, therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an

Task Description:	ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 activity (by using kinase assays) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to A1-5 cells, a rat embryo fibroblast line in which CHK1 is highly expressed. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 activity and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. This approach will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Such information will provide means for protecting mammals from HZE particles-induced killing in a short term and from HZE particles-induced carcinogenesis in a long term by activating a particular pathway (ATM or ATR-regulated checkpoints, NHEJ or HRR) depending on the genetic background of the mammals. The proposal addresses a basic research topic related to the protection of individuals from the hazards of ionizing radiation in outer space. The proposed basic research is within the framework defined by the Space Radiation Health Program, and attempts to establish the scientific basis for radiation protection of humans engaged in the exploration of outer space. It is a ground-based program that studies the hypothesis that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. We believe that our results from this proposal will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Once we understand the relationships among checkpoint activation, HRR and NHEJ, we will be able to protect cells from HZE particles-induced killing by activating the particular pathway (ATM- or ATR/CHK-regulated checkpoint, HRR or NHEJ) of the cells with different genetic backgrounds. We expect that these results will have the potential for reducing the HZE particles-induced mutagenesis and carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results from these experiments will clarify the role of ATR, one of the most important checkpoint proteins in mammalian cells, in protecting cells from ionizing radiation (IR)-induced killing.
Task Progress:	In the second year of the proposal (7/1/2004-6/30/2005), we successfully performed two separate experiments in Brookhaven National Laboratory by using the NASA beam facility there. During these experiments, we focused on the first specific aim that determines whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating homologous recombination repair (HRR). By combining with some data obtained from X-ray irradiated cells, we obtained some exciting results that provided strong evidences to support our hypothesis CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. These results are reflected in the following publications: 1.Wang, H.Y., Wang, H.C., Powell, S.N., Iliakis, G. and Wang, Y. ATR affecting cell radiosensitivity is dependent on homologous recombination repair but independent of non-homologous end joining. Cancer Res. 64: 7139-7143, 2004. 2.Wang, H.Y., Hu, B., Liu, R. and Wang, Y. CHK1 Affecting Cell Radiosensitivity is Independent of Non-homologous End Joining. Cell Cycle 4: 300-303, 2005.
Bibliography Type:	Description: (Last Updated: 07/07/2021)
Articles in Peer-reviewed Journals	Wang H, Wang H, Powell SN, Iliakis G, Wang Y. "ATR affecting cell radiosensitivity is dependent on homologous recombination repair but independent of nonhomologous end joining." Cancer Res. 2004 Oct 1;64(19):7139-43. PMID: 15466211 , Oct-2004
Articles in Peer-reviewed Journals	Wang H, Hu B, Liu R, Wang Y. "CHK1 affecting cell radiosensitivity is independent of non-homologous end joining." Cell Cycle. 2005 Feb;4(2):300-3. PMID: 15655357 , Feb-2005